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Photons as tracers of space-time curvature and mass distribution in the Universe

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*Par l'espace l'univers me comprend et m'engloutit comme un point,
par la pensée je le comprends.*

– Blaise Pascal

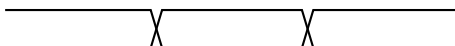
Contents

1	Introduction	1
1.1	Context	1
1.2	Outline and objectives of the project	4
2	Theoretical foundations: perturbing the homogeneous and isotropic Universe	6
2.1	The Friedmann-Lemaître-Robertson-Walker metric	6
2.2	The Christoffel symbols	8
2.3	The geodesics equations	9
2.4	The perturbed redshift	12
2.5	Overview of hypotheses	12
3	Unsheathing EXCALIBUR: Bringing Light on a Dark Universe	14
3.1	Simulation Strategy and Algorithm	14
3.2	Geometry and matter content	15
3.3	Integrator	20
3.3.1	Evaluation of numerical errors	22
4	Results and interpretation	24
4.1	Time delay measurements	27
4.2	Ongoing work	30
5	Perspectives	32
5.1	Theoretical perspectives	32
5.2	Numerical perspectives	32
5.3	Observational perspectives	33
6	Conclusion and Future Work	34
	Bibliography	36
	List of Figures	39
	List of Tables	40

CHAPTER 1

Introduction

State-of-the-art cosmological studies, more than ever, are focused on trying to unveil the nature of dark matter and dark energy. Two particular probes, two sides of a same coin, will be the focus of this Master thesis. Weak lensing (WL) and strong lensing (SL) stand as two promising avenues to probe the properties of dark matter and energy. Both are the two extreme regimes of the gravitational lensing phenomenon, and the way they are studied are very different. Those differences lie in the hypothesis that are made, in the methods themselves, and in the mathematical frameworks used to describe those phenomena. This Master thesis will stand as the first step to an attempt to reconcile the two, drawing the bridge from weak to strong lensing, by placing again the theory of General Relativity at the center of cosmological analyses. We will also discuss how we can introduce, in a single framework, many more relativistic effects related to the propagation of light such as peculiar speeds or the integrated Sachs-Wolfe effect.



1.1 Context

"Spacetime tells matter how to move, matter tells spacetime how to curve.", this quote from John A. Wheeler is often used by lecturers to summarize what the theory of General Relativity states. However, it is somewhat incomplete if one equates "matter" strictly with entities possessing mass. In this sense, photons lie outside of this definition, however, they are also impacted by the curvature of spacetime. From their emission to their observation, photons will follow their *geodesics*, so to say, the relativistic generalization of a straight line in a curved spacetime, or said differently, the generalization in relativity of the *free-falling particle*. In a completely homogeneous and isotropic Universe, these geodesics would simply be straight lines. This is logical, since the particles following those geodesics wouldn't meet any significant gravitational potential well in their path.

In a realistic description of the Universe, one starts by noting that the homogeneity and isotropy hypothesis is valid on very large scales only. Indeed, in late-time cosmology, structures have formed, and observations, especially gravitational lensing observations, show

strong tensions against those hypotheses. Strong lensing (SL) and weak lensing (WL) events would not happen in such conditions, as the path followed by photons would be unperturbed. This is where lies all the current work on analysing SL and WL events, in an attempt to map the structures of our Universe, and to understand the fundamental properties of the cosmological distributions of mass densities.

Strong lensing events (an example is showcased in figure 1.1) require a very specific alignment between the source, the lens and the observer. Weak lensing is very sensible to the clustering of matter, which allows for the differentiation between dark energy and modified gravity models. Said simply, weak lensing allows us to build a foreground galaxies map, which we can then use to infer a map of the mass distribution in the Universe. This mass map brings insights towards understanding the large-scale structure of the cosmos and for testing cosmological models. By combining weak lensing data with other probes, such as baryon acoustic oscillations or supernovae, we can place tighter constraints on the parameters governing the matter-energy contents and geometry of the Universe.

On one hand, strong lensing provides a unique opportunity to study the properties of individual galaxies or galaxy clusters. When a strong lensing event occurs, the light from a background source, such as a distant galaxy or quasar, is bent by the gravitational field of a massive foreground object, creating multiple images, arcs, or even Einstein rings. These features allow us to probe the mass distribution of the lensing object with high precision, including its dark matter content and substructure.

Strong lensing also enables the measurement of cosmological parameters, such as the Hubble constant, through time-delay cosmography. By analyzing the time delays between the multiple images of a variable source (like quasars or supernovae), we can infer the geometry of the Universe and the expansion rate ([1]). Furthermore, strong lensing can be used to detect and study faint or otherwise undetectable objects, such as quasars or dark matter clumps ([2], [3]), by observing their perturbative effects on the lensed images.

The upcoming Vera C. Rubin Observatory and its LSST (Legacy Survey of Space and Time) will revolutionize our observational capacity regarding both weak and strong lensing. With its unprecedented depth (around 24.5 magnitudes in the r-band), wide field of view (9.6 squared degrees, about 40 times the area of the full Moon, for a total of 18,000 square degrees imaged repeatedly during 10 years), and temporal resolution (each field will be visited twice every 3 nights, 15 seconds exposure each time), LSST is expected to detect millions of weak lensing shapes and tens of thousands of strong lensing events, a number around two orders of magnitude greater than current surveys. This massive increase in observational data will allow cosmologists to push lensing analyses to new levels of statistical power and precision.

In particular, the vast catalog of strong lenses expected from LSST opens the door to population-level studies of galaxy and cluster-scale lenses, as well as the possibility of identifying rare or exotic lensing configurations. Combined with LSST's deep imaging, the weak lensing shear fields will be mapped with an accuracy never reached before, over a large fraction of the sky. This dual capability — the simultaneous and coherent detection of both weak and strong lensing phenomena — is a unique opportunity to rethink the way we model gravitational lensing.

Traditionally, weak and strong lensing are studied using very different mathematical and

numerical frameworks. Weak lensing relies on perturbative treatments of the light deflection, using statistical measurements of galaxy shape distortions, and assumes that the deflection angles are small and can be linearly related to the projected potential. Strong lensing, on the other hand, requires solving non-linear lens equations and usually relies on parametric or pixel-based reconstruction of the lens mass distribution.

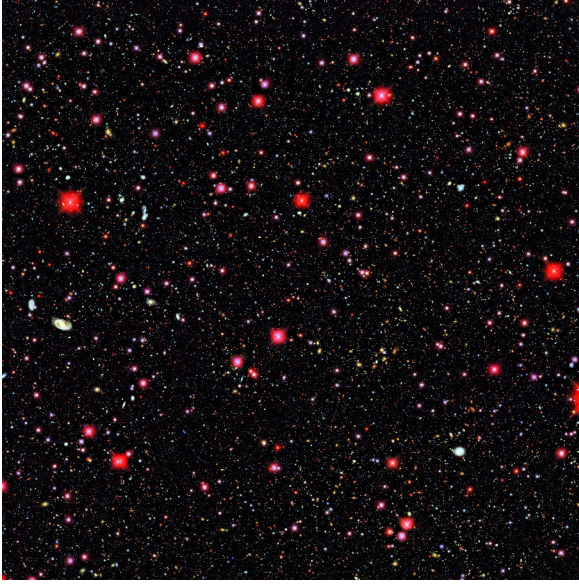
These frameworks, while efficient in their respective domains, often feel disconnected, as if they described two entirely different phenomena. However, they are both manifestations of the same underlying principle: photons following null geodesics in a curved spacetime. This fundamental idea, rooted in the heart of General Relativity, offers a natural unifying formalism.

Starting again from the roots of General Relativity will allow us not only to study gravitational lensing, but to add all the additional relativistic effects that are related to the propagation of light through curved space-times. For instance, it will be natural and straightforward to look at the effect of moving lenses on redshifts and lensing events. It will also be possible, down the line, to study different solutions to the Einstein's field equation, such as the Szekeres solution, and see what insights on cosmology we can gather from the propagation of light rays through this spacetime [4]. This is a great advantage of this method, as those effects are generally studied by independant formalisms and communities.

The approach developed in this thesis: computing lensing effects directly from the integration of null geodesics equations in a perturbed spacetime metric, seeks to establish this bridge between weak and strong lensing regimes. It allows us to study continuous deformations of geodesics under varying gravitational potentials, without the need to distinguish a priori between "weak" or "strong" lensing. This could prove especially useful in intermediate regimes, such as those that will be hopefully more common in LSST data: systems exhibiting both weak distortions on large scales and strong lensing features near the center.

Such a framework could eventually lead to a more coherent modeling of light propagation in realistic cosmological environments, capturing small and large-scale structures, line-of-sight effects, and relativistic corrections in a unified treatment. It also paves the way for new simulations and inversion tools, capable of constraining the mass distribution and cosmological parameters from lensing observations across all regimes.

In a time where precision cosmology demands consistency and global modeling strategies, this approach represents a step toward a geometrically consistent, physically grounded, and computationally robust formalism for gravitational lensing. With LSST and other missions like Euclid and the Roman Space Telescope, the potential of such unified lensing analyses is vast and largely untapped.



(a) The image below, from the Deep Lens Survey (DLS), gives a taste of what LSST skies will look like. It covers about half a square degree, and reveals ten times more galaxies per unit area than the SDSS. LSST will go even deeper.



(b) In this Hubble image of galaxy cluster SDSS J1038+4849, two bright galaxies form the "eyes," and the arcs (the "smile") result from weak and strong gravitational lensing.

Figure 1.1: Examples of images from previous galaxy surveys : (a) shows the density of background galaxies LSST will capture for weak lensing analysis. (b) shows a spectacular strong lensing event caused by a massive foreground galaxy cluster.

1.2 Outline and objectives of the project

As stated above, current cosmological analysis of weak and strong lensing events are based on very different mathematical frameworks. The objective of this Master thesis is to establish a bridge between the two, by using the same mathematical framework to describe both phenomena. This will be done thanks to the use of a very well known theory in cosmology, the theory of General Relativity. Briefly, the idea is to use the metric describing an homogeneous and isotropic spacetime, and to perturb it to the first order. Such perturbations will propagate through the Christoffel symbols, which will then be used to compute the geodesics of the perturbed metric. Those sets of equations, implemented in a correct way, and accompanied by the right numerical methods, will allow us to compute the path followed by photons in a perturbed Universe, in a closer-to-reality approach. Once the photon paths are obtained, we have the possibility to compute the lensing effects we want, be it weak or strong lensing. This will allow to study the deformations caused by cosmic structures, while also keeping an eye open and set on some of the most important cosmological quantities, such as redshifts, which are often not included in lensing studies. The project will be divided into two main parts. The first part will be dedicated to the theoretical aspects of the project, and will be based on the theory of General Relativity. The second part will be dedicated to the numerical

implementation of the theoretical study, and will be based on the use of Python, it will serve as showcasing for the final code we aim to develop, which will be written in C++ for performance reasons. Ultimately, the goal of the project is to develop a code that will be able to compute the lensing effects in a perturbed Universe, and to compare the results with the results obtained from the current methods used in cosmology, such as current weak and strong lensing analysis, for instance. Simply put, we want to do study cosmology with galaxy clusters [5], in a completely relativistic way. The code will be developed in a modular way, so that it can be easily adapted to different cosmological models and different lensing effects. The code will be made publicly available, so that it can be used by other researchers in the field. Of course, this project inscribes itself in a much larger timeframe, as this thesis is only redacted after the first two months of an internship, and the project will be continued as part of a PhD thesis. The goal of this Master thesis is to provide a solid foundation for the continuation of the project. This work inscribes itself as the continuation of a previous study [6] with the idea of generalizing ideas to more relativistic effects and with the objective of measuring masses and shapes of dark matter halos accurately. It will be useful to compare results with the work of M.-A. Breton and V. Reverdy [7], and to go further down the line by studying more localized lensing effects as well as applying the procedure to different simulation codes, other than RAMSES [8]. This work is motivated by previous articles such as H. Macpherson's [9] estimations of relativistic effects on cosmological distances measurements. Also, we could try to implement a simple way of studying light propagation under different metrics, so to compare with other raytracing codes, such as GYOTO [10].

CHAPTER 2

Theoretical foundations: perturbing the homogeneous and isotropic Universe

2.1 The Friedmann-Lemaître-Robertson-Walker metric

The Friedmann-Lemaître-Robertson-Walker (FLRW) metric is a solution to Einstein's field equations in General Relativity that describes a homogeneous and isotropic Universe. It is commonly used in cosmology to model the large-scale structure of the Universe. In spherical coordinates, the FLRW metric can be written as:

$$ds^2 = -c^2 dt^2 + a^2(t) \left(\frac{1}{1 - kr^2} dr^2 + r^2 d\Omega^2 \right) \quad (2.1)$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$ is the angular part of the metric, and $k = 0, 1, -1$ corresponds to flat, closed and open Universe respectively. The scale factor $a(t)$ describes the expansion of the Universe, so to say, how physical lengths evolve through time. The metric is homogeneous and isotropic, meaning that it looks the same in all directions and at all points in space, this appears explicitly in the metric, as there are no dependencies on the angles θ and ϕ . As stated above, the homogeneity and isotropy hypothesis is a good approximation on large scales, but on smaller scales, the Universe is inhomogeneous and anisotropic due to the presence of galaxies, clusters of galaxies, and other structures.

As such, the FLRW metric will be used as a background metric, which will be subject to perturbations. The perturbations will be treated as small deviations from the FLRW metric. The total metric and its inverse can be written as:

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu} \quad (2.2)$$

$$g^{\mu\nu} = \bar{g}^{\mu\nu} - h^{\mu\nu} + \mathcal{O}(h^2) \quad (2.3)$$

where $\bar{g}_{\mu\nu}$ is the background metric, and $h_{\mu\nu}$ is the perturbation. The perturbations can be decomposed into scalar, vector, and tensor perturbations. The scalar perturbations are the most common and are related to density fluctuations in the Universe. Those are the perturbations we are the most interested in, as they represent the gravitational potentials.

The vector terms are associated with vorticity and shear, while the tensor terms are related to gravitational waves. The perturbations can be written in a gauge-invariant way, which means that they are independent of the choice of coordinates. In the Newtonian gauge, those perturbations can be written in terms of the metric perturbations as

$$h_{\mu\nu} = \begin{pmatrix} -2\phi & -c \cdot a(t) B_{|i} \\ -c \cdot a(t) B_{|j} & 2a^2(t) \left(-\frac{\psi}{c^2} \gamma_{ij} + E_{|ij} \right) \end{pmatrix} \quad (2.4)$$

$$\bar{g}_{\mu\nu} = \begin{pmatrix} -c^2 & 0 \\ 0 & a^2(t) \gamma_{ij} \end{pmatrix} \quad (2.5)$$

which gives, when combined with the background metric, the perturbed FLRW metric

$$ds^2 = -c^2(1 + 2\phi/c^2)dt^2 - 2c \cdot a(t) B_{|i} dx^i dt - a^2(t) [(1 - 2\psi)\gamma_{ij} + 2E_{|ij}] dx^i dx^j \quad (2.6)$$

In the decomposition of metric perturbations, ϕ and ψ are the scalar potentials, B_i is the (transverse) vector perturbation, and E_{ij} is the tensor perturbation. Note that E_{ij} is a symmetric, traceless 3D tensor that can contain scalar, vector, and tensor components. However, in the conformal Newtonian gauge, the scalar and vector parts of E_{ij} are set to zero — the former by gauge choice, and the latter because it decays rapidly in an expanding Universe ([11]). The notation $x_{|i}$ indicates a covariant derivative with respect to the spatial metric γ_{ij} . The perturbations are assumed to be small, so that all equations can be linearized around $\bar{g}_{\mu\nu}$. Before moving on, there are two points we need to address.

Firstly, as mentioned above, there are three types of perturbations: scalar, vector and tensor. All computations done in this master thesis are done using the whole three types of perturbations, however, given the short timeframe in which this work is inscribed, we will consider here only scalar perturbations. This means the Christoffel symbols and geodesic equations are obtained with all the perturbations terms and for a k -curved space, then they are set to zero, as well as the curvature. This is a valid approximation, as the vector perturbations vanish early on in the evolution of the Universe, and the tensor perturbations are negligible in a cosmological setting, compared to the scalar perturbations. It will be interesting to see how some of the results may be altered by the presence of gravitational waves, for instance.

The second point is about the further numerical treatment of the project. In numerical simulations, it is way more convenient to work with cartesian coordinates than spherical coordinates. Defining grids and evaluating gradients and/or potentials is way more straightforward this way, and most current cosmological simulations are done in cartesian coordinates. However, the FLRW metric is defined in spherical coordinates. This means that we need to operate a change of coordinates from spherical to cartesian coordinates. Recalling eq. (2.1), we use the following change of coordinates:

$$r = \frac{\rho}{1 + \frac{k}{4}\rho^2} \quad (2.7)$$

which allows us to rewrite the metric as

$$ds^2 = -c^2 dt^2 + \frac{a^2(t)}{(1 + k\rho^2/4)^2} (d\rho^2 + \rho^2 d\Omega^2) \quad (2.8)$$

We recognize the term $d\rho^2 + \rho^2 d\Omega^2$ as the flat metric in cartesian coordinates. This means that we can finally write the FLRW metric in cartesian coordinates

$$ds^2 = -c^2 dt^2 + \frac{a^2(t)}{[1 + k(x^2 + y^2 + z^2)/4]^2} (dx^2 + dy^2 + dz^2) \quad (2.9)$$

which will make the computations of the Christoffel symbols and the geodesic equations easier, along with the numerical treatment of the overall problem.

2.2 The Christoffel symbols

The Christoffel symbols are a set of functions that describe how the coordinates of a manifold change when we move from one point to another. They are defined in terms of the metric tensor and its derivatives. The Christoffel symbols are given by:

$$\Gamma_{\alpha\beta}^{\mu} = \frac{1}{2} g^{\mu\nu} (\partial_{\alpha} g_{\nu\beta} + \partial_{\beta} g_{\nu\alpha} - \partial_{\nu} g_{\alpha\beta}) \quad (2.10)$$

where $g^{\mu\nu}$ is the inverse of the metric tensor $g_{\mu\nu}$, and ∂_{α} denotes the partial derivative with respect to the coordinate x^{α} . It is important to note that the Christoffel symbols are not tensors themselves, but they are used to define the covariant derivative, which is a tensorial object.

We are interested in the perturbed Christoffel symbols, which can be obtained by substituting the perturbed metric (2.3) into eq. (2.10). The perturbed Christoffel symbols can be written as:

$$\Gamma_{\alpha\beta}^{\mu} = \frac{1}{2} (\bar{g}^{\mu\nu} - h^{\mu\nu}) (\partial_{\alpha} (\bar{g}_{\nu\beta} + h_{\nu\beta}) + \partial_{\beta} (\bar{g}_{\nu\alpha} + h_{\nu\alpha}) - \partial_{\nu} (\bar{g}_{\alpha\beta} + h_{\alpha\beta})) \quad (2.11)$$

Developing this expression, neglecting terms of order h^2 , one obtains the perturbed Christoffel symbols, which can be written as:

$$\Gamma_{\alpha\beta}^{\mu} = \bar{\Gamma}_{\alpha\beta}^{\mu} + \delta\Gamma_{\alpha\beta}^{\mu} \quad (2.12)$$

where $\bar{\Gamma}_{\alpha\beta}^{\mu}$ are the Christoffel symbols of the background metric, and $\delta\Gamma_{\alpha\beta}^{\mu}$ are the perturbations to the Christoffel symbols. The perturbations can be written as:

$$\delta\Gamma_{\alpha\beta}^{\mu} = \frac{1}{2} \bar{g}^{\mu\nu} (\partial_{\alpha} h_{\beta\nu} + \partial_{\beta} h_{\alpha\nu} - \partial_{\nu} h_{\alpha\beta}) - \frac{1}{2} h^{\mu\nu} (\partial_{\alpha} \bar{g}_{\beta\nu} + \partial_{\beta} \bar{g}_{\alpha\nu} - \partial_{\nu} \bar{g}_{\alpha\beta}) \quad (2.13)$$

The final expression for the perturbed Christoffel symbols is

$$\begin{aligned}\Gamma_{\alpha\beta}^{\mu} = & \frac{1}{2}\bar{g}^{\mu\nu}(\partial_{\alpha}\bar{g}_{\nu\beta} + \partial_{\beta}\bar{g}_{\nu\alpha} - \partial_{\nu}\bar{g}_{\alpha\beta}) \\ & + \frac{1}{2}\bar{g}^{\mu\nu}(\partial_{\alpha}h_{\beta\nu} + \partial_{\beta}h_{\alpha\nu} - \partial_{\nu}h_{\alpha\beta}) - \frac{1}{2}h^{\mu\nu}(\partial_{\alpha}\bar{g}_{\beta\nu} + \partial_{\beta}\bar{g}_{\alpha\nu} - \partial_{\nu}\bar{g}_{\alpha\beta})\end{aligned}\quad (2.14)$$

This is the expression we will use to compute the Christoffel symbols that are involved in the geodesic equations. The first term is the background Christoffel symbols, which are the Christoffel symbols of the unperturbed FLRW metric (2.1). The second and third terms are the perturbations to the Christoffel symbols. In a $k = 0$ spacetime, and with all vector and tensor perturbations set to zero, the Christoffel symbols are given by:

Γ_{00}^0	$\frac{1}{c^2} \frac{\partial \Phi}{\partial t}$
Γ_{ii}^i	$-\frac{1}{c^2} \frac{\partial \Psi}{\partial x^i}$
Γ_{ii}^0	$\frac{a\dot{a}}{c^2} + \frac{2a\dot{a}}{c^4}(\Phi + \Psi) - \frac{a^2}{c^4} \frac{\partial \Psi}{\partial t}$
Γ_{00}^i	$\frac{1}{a^2} \frac{\partial \Phi}{\partial x^i}$
$\Gamma_{0i}^0 = \Gamma_{i0}^0$	$\frac{1}{c^2} \frac{\partial \Phi}{\partial x^i}$
$\Gamma_{i0}^i = \Gamma_{0i}^i$	$\frac{\dot{a}}{a} - \frac{1}{c^2} \frac{\partial \Psi}{\partial t}$
Γ_{jj}^i	$\frac{1}{c^2} \frac{\partial \Psi}{\partial x^i}$
$\Gamma_{ij}^i = \Gamma_{ji}^i$	$-\frac{1}{c^2} \frac{\partial \Psi}{\partial x^j}$

Table 2.1: The perturbed Christoffel symbols in a $k = 0$ spacetime, with all vector and tensor perturbations set to zero. The terms Φ and Ψ are the scalar perturbations. The terms $\dot{a}$ and a are the time derivative of the scale factor and the scale factor respectively.

2.3 The geodesics equations

In General Relativity, the motion of a free particle is described by geodesics, which are the solutions to the geodesic equations. They are derived from the principle of least action, which states that the path taken by a particle is the one that minimizes the action. We are interested in those, as they will describe the deformation of the path of photons in the vicinity of gravitational potentials. The geodesic equation can be written as:

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0 \quad (2.15)$$

where λ is an affine parameter along the geodesic, and $\Gamma_{\alpha\beta}^\mu$ are the Christoffel symbols, which are functions of the metric and its derivatives. The geodesic equation describes how the position of a particle changes with respect to the affine parameter. The affine parameter is a free parameter, however, it is often chosen to be the proper time τ for massive particles. On the other hand, photons are massless, their structure in General Relativity forbids them from having a proper time. This means that the numerical treatment of this parameter needs to be discussed, which will be done later on.

Eq. (2.15) needs to be written in terms of the perturbed metric, which can be done by substituting the perturbed metric (2.6) into the geodesic equation, more precisely into the Christoffel symbols. Replacing $\Gamma_{\alpha\beta}^\mu$ by eq. (2.14) in eq. (2.15) gives:

$$\frac{d^2 x^\mu}{d\lambda^2} + \left(\bar{\Gamma}_{\alpha\beta}^\mu + \delta\Gamma_{\alpha\beta}^\mu \right) \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0 \quad (2.16)$$

Using the results from Table 2.1, we derive the four geodesic equations for the perturbed FLRW metric.

$$\begin{aligned} \frac{d^2 t}{d\lambda^2} + \frac{1}{c^4} \left[\left(\left(\frac{dx}{d\lambda} \right)^2 + \left(\frac{dy}{d\lambda} \right)^2 + \left(\frac{dz}{d\lambda} \right)^2 \right) \left(a(t)\dot{a}(t)(c^2 - 2\Phi - 2\Psi) - a(t)^2 \frac{\partial\Psi}{\partial t} \right) \right. \\ \left. + c^2 \frac{dt}{d\lambda} \left(\frac{dt}{d\lambda} \frac{\partial\Phi}{\partial t} + 2 \frac{dx}{d\lambda} \frac{\partial\Phi}{\partial x} + 2 \frac{dy}{d\lambda} \frac{\partial\Phi}{\partial y} + 2 \frac{dz}{d\lambda} \frac{\partial\Phi}{\partial z} \right) \right] = 0 \end{aligned} \quad (2.17)$$

$$\begin{aligned} \frac{d^2 x}{d\lambda^2} + \frac{1}{a(t)^2} \left(\frac{dt}{d\lambda} \right)^2 \frac{\partial\Phi}{\partial x} + \frac{2\dot{a}(t)}{a(t)} \frac{dt}{d\lambda} \frac{dx}{d\lambda} \\ + \frac{1}{c^2} \left[-2 \frac{dx}{d\lambda} \left(\frac{dt}{d\lambda} \frac{\partial\Psi}{\partial t} + \frac{dy}{d\lambda} \frac{\partial\Psi}{\partial y} + \frac{dz}{d\lambda} \frac{\partial\Psi}{\partial z} \right) \right. \\ \left. - \left(\frac{dx}{d\lambda} \right)^2 \frac{\partial\Psi}{\partial x} + \left(\left(\frac{dy}{d\lambda} \right)^2 + \left(\frac{dz}{d\lambda} \right)^2 \right) \frac{\partial\Psi}{\partial x} \right] = 0 \end{aligned} \quad (2.18)$$

$$\begin{aligned} \frac{d^2 y}{d\lambda^2} + \frac{1}{a(t)^2} \left(\frac{dt}{d\lambda} \right)^2 \frac{\partial\Phi}{\partial y} + \frac{2\dot{a}(t)}{a(t)} \frac{dt}{d\lambda} \frac{dy}{d\lambda} \\ + \frac{1}{c^2} \left[-2 \frac{dy}{d\lambda} \left(\frac{dt}{d\lambda} \frac{\partial\Psi}{\partial t} + \frac{dx}{d\lambda} \frac{\partial\Psi}{\partial x} + \frac{dz}{d\lambda} \frac{\partial\Psi}{\partial z} \right) \right. \\ \left. - \left(\frac{dy}{d\lambda} \right)^2 \frac{\partial\Psi}{\partial y} + \left(\left(\frac{dx}{d\lambda} \right)^2 + \left(\frac{dz}{d\lambda} \right)^2 \right) \frac{\partial\Psi}{\partial y} \right] = 0 \end{aligned} \quad (2.19)$$

$$\begin{aligned}
 & \frac{d^2 z}{d\lambda^2} + \frac{1}{a(t)^2} \left(\frac{dt}{d\lambda} \right)^2 \frac{\partial \Phi}{\partial z} + \frac{2\dot{a}(t)}{a(t)} \frac{dt}{d\lambda} \frac{dz}{d\lambda} \\
 & + \frac{1}{c^2} \left[-2 \frac{dz}{d\lambda} \left(\frac{dt}{d\lambda} \frac{\partial \Psi}{\partial t} + \frac{dx}{d\lambda} \frac{\partial \Psi}{\partial x} + \frac{dy}{d\lambda} \frac{\partial \Psi}{\partial y} \right) \right. \\
 & \left. - \left(\frac{dz}{d\lambda} \right)^2 \frac{\partial \Psi}{\partial z} + \left(\left(\frac{dx}{d\lambda} \right)^2 + \left(\frac{dy}{d\lambda} \right)^2 \right) \frac{\partial \Psi}{\partial z} \right] = 0
 \end{aligned} \tag{2.20}$$

Those equations describe the motion of a photon in a flat perturbed FLRW metric, containing only scalar perturbations, equivalently, gravitational potentials. The equations are coupled, and they need to be solved simultaneously. They are second order differential equations, which means that they need initial positions and velocities to be solved. However, an attentive eye might have noticed that nowhere during the derivation of the geodesic equations have we specified in some way that the particle we are interested in is a photon. This means that the equations we have derived are valid for any particle, massive or massless. Of course, geodesics are not supposed to be the same for massive and massless particles. Therefore, we need a fifth equation, which will act as a constraint on the four geodesic equations. A particular condition, proper to photons, comes to mind :

$$k_\mu k^\mu = \bar{g}_{\mu\nu} k^\mu k^\nu = 0 \tag{2.21}$$

which we recall, is the definition of a light-like 4-vector in General Relativity. Developing using the background metric (2.9) gives:

$$\left(\frac{dt}{d\lambda} \right)^2 = \frac{a^2(t)}{c^2(1 + k(x^2 + y^2 + z^2)/4)^2} \left(\left(\frac{dx}{d\lambda} \right)^2 + \left(\frac{dy}{d\lambda} \right)^2 + \left(\frac{dz}{d\lambda} \right)^2 \right) \tag{2.22}$$

We recall that for now, we are working in a flat Universe, so $k = 0$. This means that the equation (2.22) simplifies to::

$$\left(\frac{dt}{d\lambda} \right)^2 = \frac{a^2(t)}{c^2} \left(\left(\frac{dx}{d\lambda} \right)^2 + \left(\frac{dy}{d\lambda} \right)^2 + \left(\frac{dz}{d\lambda} \right)^2 \right) \tag{2.23}$$

Numerically, this means that we can, in some sense, completely forget about the time-component geodesic, only work with the spatial geodesics and use them as a normalization of the time speed component. This also means that we go from a set of 4 equations to a set of 3 equations, which is an elegant conclusion to the problematic of describing the path of photons.

2.4 The perturbed redshift

As explained in the next section, this result will not be useful for the rest of this report, however we still want to be exhaustive and give it, as it will serve as grounds for the continuation of the internship. For reasons of length, we will not delve into the details of the calculations. Redshift is defined as follows

$$z = \frac{(g_{\mu\nu}k^\mu u^\nu)_S}{(g_{\mu\nu}k^\mu u^\nu)_O} - 1 \quad (2.24)$$

with $k^\mu = \left(\frac{dt}{d\lambda}, \frac{dx}{d\lambda}, \frac{dy}{d\lambda}, \frac{dz}{d\lambda}\right)$ and u^μ the 4-velocity of the observer. When considering the perturbations of the energy-momentum tensor $T_{\mu\nu}$, and the normalization conditions on the 4-velocity $u_\mu u^\mu = -1$, we get that [6], [12]

$$1 + z = \frac{a(t_O)}{a(t_S)} \left[1 + (\Phi_S - \Phi_O) + (\vec{n} \cdot \vec{v}_O - \vec{n} \cdot \vec{v}_S) - 2 \int_{\lambda_S}^{\lambda_O} \frac{\partial \Phi}{\partial \eta} d\lambda \right] \quad (2.25)$$

2.5 Overview of hypotheses

Before we go on to the numerical integration of the geodesic equations, let us lay down the hypothesis already made, the ones we will make and the ones we will not make.

- We will work in a flat Universe, $k = 0$. The treatment of a cosmological simulation in curved space is still an open question in the numerical cosmology community. This is the most classical case of cosmological simulations.
- We will work with scalar perturbations only. This means that we will neglect vector and tensor perturbations. This is a good approximation, as scalar perturbations are the dominant ones in the current Universe. However, we will have a look at the consequences of a stochastic background of gravitational waves on the deformation of lighttrays fairly early in the future days of this project.
- Given the short timeframe of this project, and especially of this report, we will not take into account the expansion of the Universe. This means that we will work in a static Universe. This raises many questions, especially regarding redshifts, since this completely forfeits the cosmological redshift. Also regarding visualization, as the Universe is not expanding, the light rays can not be visualized the same way as in a real cosmological simulation. However, this is a good first exercise, and will allow us to make the first steps in the right direction. We still have the opportunity to visualize 3D trajectories of photons in a static Universe without having to worry about their representation under the expansion of the Universe. We will not be able to take redshifts into account as we consider a static Universe without any peculiar velocities for now. Also, since sources are not determined and we are first solely interested in the propagation of photons, referring to eq. (2.25), all terms vanish except for the difference of potentials. This term holds no physical significance in our case as the observer is fixed

and they are no sources. Obviously, taking into account the expansion of the Universe will be the next step and priority of the project.

The approach we adopt towards generating gravitational lensing in a cosmological simulation, which is to use General Relativity in its most direct application, will allow us to get rid of some approximations usually made in lensing analysis. The Born approximation [13] is one of them, it is a good approximation for weak gravitational fields, and for photons that are not too close to the lens. This means that under this hypothesis, we can not take go from weak lensing modeling to strong lensing modeling. This is a problem, as we are interested in strong lensing, and the intermediate regimes between weak and strong lensing. The Born approximation consists in neglecting the deflection angle of the light rays when evaluating the gravitational potential. This can prove limiting when considering higher orders of deformations in weak lensing, such as flexion, for instance. To summarize, in weak lensing analysis, the following hypothesis are made:

- The Born approximation: evaluating deflections with respect to the straight trajectory.
- Stationary potentials: the gravitational potential is not time-dependent.
- Thin lens approximation: massive structures are modelled as thin planes, located at a given redshift. This means that the lens effect is instant once the light ray crosses the lens plane.
- Single lens plane: only one lens plane is considered at a time, which forfeits completely multilens related effects and non linear propagation effects between multiple structures.

Generating gravitational lensing using General Relativity will allow us to instantly drop all of those approximations. This will allow us to shed light on some inconsistencies between the different approaches.

CHAPTER 3

Unsheathing EXCALIBUR: Bringing Light on a Dark Universe

We now present EXCALIBUR, which stands for **EX**act **CAL**culation of **L**ight **B**ending Using **R**elativity. While the name is a reference to the legendary sword of King Arthur, it also alludes to the fact that EXCALIBUR is a tool that will, in the most literal sense, *shed light* through a given cosmological setting, using nothing more than General Relativity. The tool is intended, in the future, to be written in C++ and to use the full power of this programming language in regards to speed and efficiency. However, for the sake of this master thesis, we will present a prototype in Python, which is a more accessible language for demonstrating the algorithm and its results. The code is available at <https://github.com/lmagristella/excalibur>.

3.1 Simulation Strategy and Algorithm

Let us use a bottom-up approach to explain the algorithm and numerical framework we need to implement in order to integrate the geodesic equations. Taking a look at eqs. (2.17) to (2.20), we can see that we need to be able to evaluate, at any given point in spacetime, the potential Φ and the gradient of the potential $\nabla\Phi$. For the sake of generality, we don't want to simply assume that all dark matter halos are spherical. This means that if we are interested in some sorts of elliptical halos, we need to be able to numerically evaluate the potential and its gradient without needing an analytical expression. In return, this means that we shouldn't have to call some methods to evaluate them. What we mean by that is that the potentials and gradients emanating from massive objects should be precomputed before hand, and stored in a grid, especially when considering multiple structures that are not necessarily spherical.

Let us note that once again, due to the short timeframe of the project, we will first consider spherical masses in this master thesis. This will also allow us, in the future, to compare with usual results, that usually assume such mass shapes. It is well known that spherically symmetric mass distributions often present an analytical solution to the Poisson equation

$$\vec{\nabla}^2\Phi = 4\pi G\rho \tag{3.1}$$

such that it would be easier to just evaluate the related solution of that equation at each point in the trajectory of the photon. However, as aforementioned, we want our approach to be directly adaptable to unusual and not necessarily analytically solvable cases, for instance to a distribution of N ellipsoidal dark matter halos. We will therefore need to implement a grid of points in the three-dimensional space, and to evaluate the potential and its gradient at each point in the grid.

3.2 Geometry and matter content

In this section, we present how space-time and astrophysical objects are represented and handled within the code. The code is structured around two main components: one that creates and manages a three-dimensional grid of points, and another that defines the astrophysical objects to be placed within this grid. Our goal is to make the entire process, from creating the grid to defining and positioning the objects, and finally evaluating the gravitational potential and its gradient at each grid point—as straightforward and user-friendly as possible.

```
1 import numpy as np
2 from excalibur.grid import Grid3D
3 from excalibur.objects import spherical_mass
4 from excalibur.constants import *
5 from excalibur.integrator import *
6
7 N = 2**10 # 1024, resolution of the grid
8
9 # position of the observer, in normalized comoving coordinates
10 observer_position = np.array([0, 0, 0])
11 observing_angle = (np.arccos(np.sqrt(3)/3), np.pi/4) #angle of observation
12 aperture_angle = 7 * np.pi/180 # 7 degrees
13 cone = (observer_position, observing_angle, aperture_angle)
14
15 #creating a NxNxN grid of points, with a specified observation cone
16 grid = Grid3D(N, cone = cone)
```

This code snippet creates a grid of size $1024 \times 1024 \times 1024$, with the observer at the origin, observing towards the center of the grid, with an aperture angle of 7 degrees (see fig. 3.1). One particular question quickly arises: supposing that each cell of the grid contains a float64 value, how much memory would we need to store the grid? If we need to store a density value, a potential value, and a 3D vector for the gradient of the potential, we would need to store $5 \times 8 \times 1024^3 \approx 40$ GB of memory. This is a lot of memory if this is to be stored on a laptop. The obvious solution to this problem, that a numerical physicist would immediately suggest, would be to use an AMR (Adaptive Mesh Refinement) grid. This would allow us to store the grid in a more efficient way, by only creating grid points where the matter density is high enough, and by using a coarser grid where the matter density is low. However, this would also require a more complex implementation, and we will not be able to implement it

in the time frame of this project. Moreover, the benefits of using an AMR grid with Python are not as significant as with C++, due to the fact that Python is not as efficient as C++ in terms of memory management. We will therefore go to the second best solution, which is to use sparse arrays. This type of array is a data structure that only stores the non-zero values of the array, since most of the values in the grid will be zero, given that we defined a cone of observation. This will allow us to save a lot of memory, and to store the grid in a more efficient way. For instance, using the defined cone from the previous code snippet, we can estimate that we will only need to store about 1.5% of the grid, which would mean that we would only need to store about 600 MB of memory, which is much more manageable.

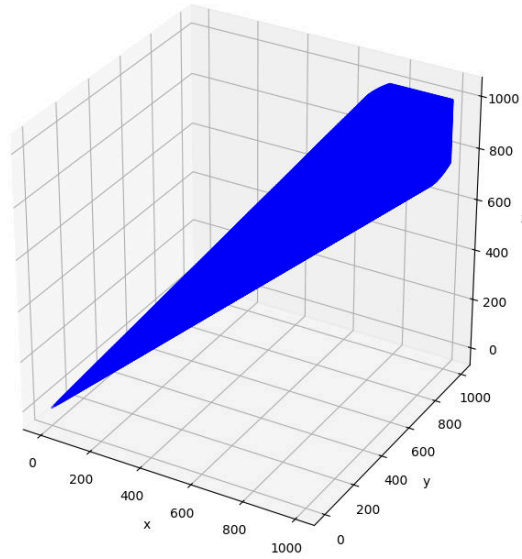


Figure 3.1: Visualizing the cone of observation mentioned earlier. The observer is located at the origin, and the cone is defined by the angle of observation and the aperture angle. The cone is represented by the blue points.

Let us now place some objects in the grid.

```

1 M = 1e13 # mass of the halo, in Msun
2 R = 1 # radius of the halo, in Mpc
3 center = np.array([0.5, 0.5, 0.5]) # position of the center of the halo
4
5 halo = spherical_mass(M, R, center=center, density_profile='uniform')

```

Here we have defined a spherical halo with a mass of $10^{13}M_{\odot}$ and a radius of 1 Mpc, with a uniform density profile. The mass and density profile are not really realistic but this is a first approach towards more generic descriptions of spherical masses. The mass is a bit low, while usual halos range around $10^{14}/10^{15}M_{\odot}$, and their density profiles are not at all uniform. We can now place this halo in the grid, and evaluate the potential and its gradient at each point in the grid.

```
1 grid.set_phi(halo.potential)
2 grid.set_density(halo.density)
3 grid.compute_gradients()
```

We have now assigned to every cell of the grid the potential and density values, along with the gradient vector of the potential.

On a related note, the grid can be saved to a .npz file, which is a compressed numpy format that allows us to save the grid in a more efficient way. This allows us to save time and memory, and to create many different grids with different parameters without having to recompute the grid every time. We only need to load them, and we can then use them to compute the geodesics.

Once the grid is created, and the relevant physical quantities are computed for all the cells of the grid, we can try to visualize some slices of the grid to see what it looks like.

```
1 import matplotlib.pyplot as plt
2
3 fig, ax = plt.subplots(1, 2, figsize=(12, 5))
4 if grid.is_sparse:
5     slice_img = grid.get_slice(axis='z', value=0.5).todense()
6 else:
7     slice_img = grid.get_slice(axis='z', value=0.5)
8
9 im = ax[0].imshow(slice_img, origin='lower', cmap='viridis')
10 fig.colorbar(im, ax=ax[0], label=r'$\phi$'+' in Mpc2/Gyr2')
11 ax[0].set_xlabel('x')
12 ax[0].set_ylabel('y')
13 ax[0].set_title(f'Slice de $\phi$ pour z=0.5')
14 if grid.is_sparse:
15     slice_img = grid.get_slice(axis='z',
16                               value=0.5, quantity="density").todense()
17 else:
18     slice_img = grid.get_slice(axis='z', value=0.5, quantity="density")
19
20 im = ax[1].imshow(slice_img, origin='lower', extent=[0, 1, 0, 1],
21                  cmap='viridis', vmin=0, vmax=np.max(grid.rho))
22 fig.colorbar(im, ax=ax[1], label=r'$\rho$'+' in Msun/Mpc3')
23 ax[1].set_xlabel('x')
24 ax[1].set_ylabel('y')
25 ax[1].set_title(f'Slice de $\rho$ pour z=0.5')
26 plt.tight_layout()
27 plt.show()
```

The `grid.get_slice()` method allows us to get a slice of the grid at a given value of the coordinate, and to visualize it. The `quantity` argument allows us to specify which quantity we want to visualize, either the potential or the density, or a gradient component. The `axis`

argument allows us to specify which axis we want to slice, either x , y or z . There is also the option to slice the grid along a given plane, given by the normal vector and a point in the plane. The Cloud-in-Cell (CIC) method is used to interpolate the values of the grid at the given coordinates, and to assign the values to the cells of the grid. This method treats particles not as point masses, but as "clouds" with a finite volume, distributing their properties (such as mass or potential) over the nearest grid cells according to a linear weighting. It ensures a smoother and more continuous assignment compared to simpler schemes like the Nearest Grid Point (NGP) method, reducing aliasing effects and improving the accuracy of field estimations.

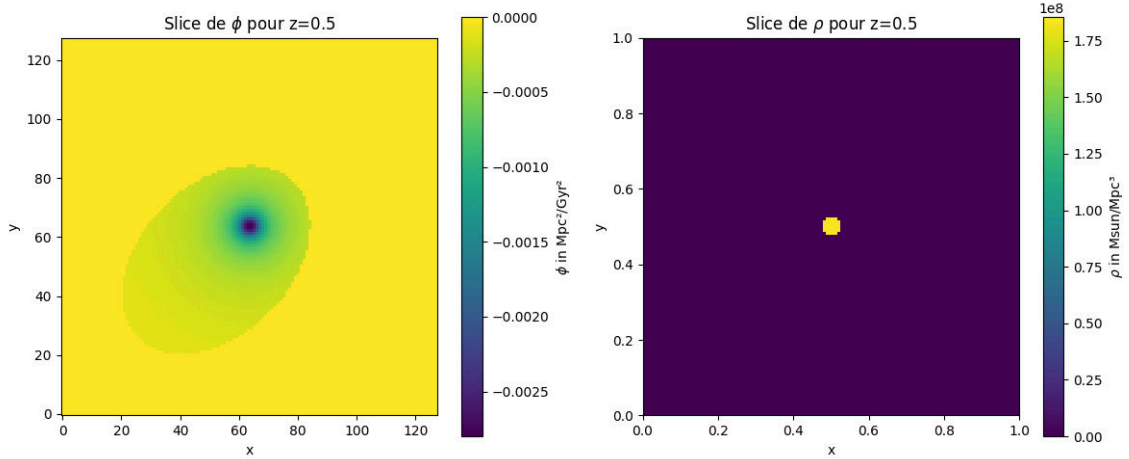


Figure 3.2: Slice of the potential (left) and density (right) of a single halo, with a mass of $10^{13}M_{\odot}$ and a radius of 1 Mpc, at coordinate $z=0.5$. The potential is represented in $\text{Mpc}^2/\text{Gyr}^2$, and the density is represented in Msun/Mpc^3 .

The left plot shows the potential of the halo, and the right plot shows the density distribution of the halo. We can explicitly see the ellipse caused by the slicing in z of the cone defined earlier, it is also present in the density plot, but it is less visible due to the fact that the density is zero outside the halo, as well as outside the cone.

The potential represented here has been computed using the analytical solution of the Poisson equation for a spherically symmetric and uniform mass distribution. However, we will want to look at more complex mass distributions, which means that we will need to compute the potential numerically, via the Poisson equation. One way to do this is to use the Fast Fourier Transform (FFT) method, which allows us to compute the potential in an efficient and general way. One comment we need to make here is that the FFT assumes that the grid is periodic. This means that we need to periodicize the grid, by adding a copy of the grid on each side of the grid. This is not a problem (if we stay in three dimensions, as we can periodicize the comoving space but not the redshift space) since we are interested in cosmological scales, and creating a grid pavement of the Universe is an approach that is often used in cosmology, such as in [13]. Finally, this means that this method will be more relevant once we start looking at cosmological simulations, and not just at a single halo.

However, we will display now some results obtained with the FFT method, in order to show

the relevancy of the method. We will use a single halo, with a mass of $10^{13} M_{\odot}$ and a radius of 1 Mpc, and compute the potential using the FFT method.

```
1 grid.set_density(halo.uniform_density)
2
3 grid.set_phi(grid.poisson_solver())
```

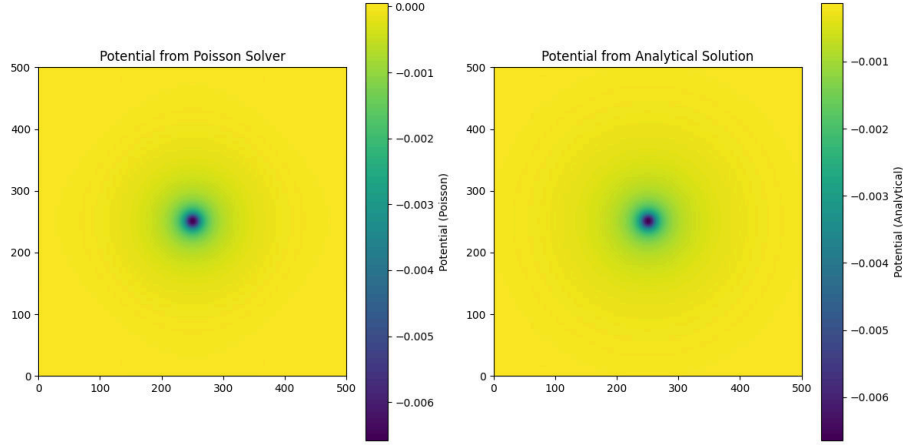


Figure 3.3: Comparing two slices of the potential, one computed using the FFT method (left) and one computed using the analytical solution (right). The potential is represented in $\text{Mpc}^2/\text{Gyr}^2$.

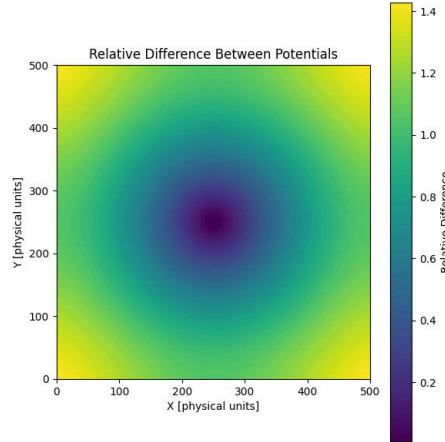


Figure 3.4: Relative difference between the potential computed using the FFT method and the potential computed using the analytical solution. The x and y axes are in Mpc. The relative difference is defined as $\frac{\phi_{fft} - \phi_{analytic}}{\phi_{analytic}}$

We can notice that the further we go from the center of the halo, the more the potential computed using the FFT method diverges from the analytical solution. This is due to the fact

we mentioned earlier, that the FFT method assumes that the grid is periodic, and as such points far away from the center of the halo will "feel" the potential of the halo coming from the other boundaries of the grid. When considering single halos, this will be somewhat unphysical, but when considering a grid of halos, this will be less of a problem, since the halos will be distributed in a way that is more analogous to the real Universe. For now, we will resort to summing the potentials of each halo at each point in the grid. This is a simple and efficient way to compute the potential of multiple halo systems, if we consider spherically symmetric distributions of mass.

3.3 Integrator

In this section, we will present the integrator that we will use to compute the geodesics of the photons. We will use a Runge-Kutta 4th order method, which is a simple and efficient method to integrate ordinary differential equations. It is precise enough for our demonstration purposes, and it is easy to implement. We will implement it from scratch, and we will not use any external library for this, as we wish to have as much control as possible over the implementation. We will lay out the general structure of the integrator, and we will then recall the set of equations we will have to solve in order to compute the geodesics. Basically, the integrator function needs to take as input a state vector, which would be a vector containing everything we need to know about the photon at a given time, and a function that will compute the derivatives of the state vector. The state vector, at the very minimum, will be a 8-dimensional vector, containing the following quantities:

$$\begin{pmatrix} t \\ x \\ y \\ z \\ \frac{dt}{d\lambda} \\ \frac{dx}{d\lambda} \\ \frac{dy}{d\lambda} \\ \frac{dz}{d\lambda} \end{pmatrix} \quad (3.2)$$

where t is the time, x , y and z are the coordinates of the photon, and λ is the affine parameter. We will add some extra quantities to the state vector, such as the redshift of the photon along the trajectory. We also can add the scale factor $a(t)$, but we will not need it for now, since we consider ourselves to be in a static Universe. The integrator is chosen to be a 4th order

Runge-Kutta method, which can be written as follows:

$$\begin{aligned}
 k_1 &= f(t_n, y_n) \\
 k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right) \\
 k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right) \\
 k_4 &= f(t_n + h, y_n + hk_3) \\
 y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)
 \end{aligned} \tag{3.3}$$

where h is the step size, t_n is the time at the beginning of the step, and y_n is the state vector at the beginning of the step. The function f is the function that computes the derivatives of the state vector. RK4 basically consists of computing the derivatives of the state vector at four different points in time, and then taking a weighted average of these derivatives to compute the state vector at the next time step. We now need to specify what the function f is. It should be a function that takes as input the state vector, and returns the derivatives of the state vector. This is nothing more than using the speed components as the derivatives of the position components, and the geodesic equations as the derivatives of the speed components. We can write it as follows:

$$f \left(\begin{pmatrix} t \\ x \\ y \\ z \\ \frac{dt}{d\lambda} \\ \frac{dx}{d\lambda} \\ \frac{dy}{d\lambda} \\ \frac{dz}{d\lambda} \end{pmatrix} \right) = \begin{pmatrix} \frac{dt}{d\lambda} \\ \frac{dx}{d\lambda} \\ \frac{dy}{d\lambda} \\ \frac{dz}{d\lambda} \\ -\frac{1}{c^4} \left(\left(\frac{dx}{d\lambda} \right)^2 + \left(\frac{dy}{d\lambda} \right)^2 + \left(\frac{dz}{d\lambda} \right)^2 \right) \left[a(t)\dot{a}(t)(c^2 - 2\Phi - 2\Psi) - a(t)^2 \frac{\partial \Psi}{\partial t} \right] + \frac{1}{c^2} \left[c^2 \frac{dt}{d\lambda} \frac{\partial \Phi}{\partial t} + 2 \frac{dx}{d\lambda} \frac{\partial \Phi}{\partial x} + 2 \frac{dy}{d\lambda} \frac{\partial \Phi}{\partial y} + 2 \frac{dz}{d\lambda} \frac{\partial \Phi}{\partial z} \right] \\ -\frac{1}{a(t)^2} \left(\frac{dt}{d\lambda} \right)^2 \frac{\partial \Phi}{\partial x} - \frac{2\dot{a}(t)}{a(t)} \frac{dt}{d\lambda} \frac{dx}{d\lambda} - \frac{1}{c^2} \left[2 \frac{dx}{d\lambda} \left(\frac{dt}{d\lambda} \frac{\partial \Psi}{\partial t} + \frac{dy}{d\lambda} \frac{\partial \Psi}{\partial y} + \frac{dz}{d\lambda} \frac{\partial \Psi}{\partial z} \right) + \left(\frac{dx}{d\lambda} \right)^2 \frac{\partial \Psi}{\partial x} - \left(\left(\frac{dy}{d\lambda} \right)^2 + \left(\frac{dz}{d\lambda} \right)^2 \right) \frac{\partial \Psi}{\partial x} \right] \\ -\frac{1}{a(t)^2} \left(\frac{dt}{d\lambda} \right)^2 \frac{\partial \Phi}{\partial y} - \frac{2\dot{a}(t)}{a(t)} \frac{dt}{d\lambda} \frac{dy}{d\lambda} - \frac{1}{c^2} \left[2 \frac{dy}{d\lambda} \left(\frac{dt}{d\lambda} \frac{\partial \Psi}{\partial t} + \frac{dx}{d\lambda} \frac{\partial \Psi}{\partial x} + \frac{dz}{d\lambda} \frac{\partial \Psi}{\partial z} \right) + \left(\frac{dy}{d\lambda} \right)^2 \frac{\partial \Psi}{\partial y} - \left(\left(\frac{dx}{d\lambda} \right)^2 + \left(\frac{dz}{d\lambda} \right)^2 \right) \frac{\partial \Psi}{\partial y} \right] \\ -\frac{1}{a(t)^2} \left(\frac{dt}{d\lambda} \right)^2 \frac{\partial \Phi}{\partial z} - \frac{2\dot{a}(t)}{a(t)} \frac{dt}{d\lambda} \frac{dz}{d\lambda} - \frac{1}{c^2} \left[2 \frac{dz}{d\lambda} \left(\frac{dt}{d\lambda} \frac{\partial \Psi}{\partial t} + \frac{dx}{d\lambda} \frac{\partial \Psi}{\partial x} + \frac{dy}{d\lambda} \frac{\partial \Psi}{\partial y} \right) + \left(\frac{dz}{d\lambda} \right)^2 \frac{\partial \Psi}{\partial z} - \left(\left(\frac{dx}{d\lambda} \right)^2 + \left(\frac{dy}{d\lambda} \right)^2 \right) \frac{\partial \Psi}{\partial z} \right] \end{pmatrix} \tag{3.4}$$

Before moving on to the next subsection, let us explain why we implemented the RK4 scheme ourselves, as Python's ecosystem proposes lots of alternatives for numerical integration. This choice was motivated by two main reasons. First, the structure of our problem is non-standard: at each integration step, the gravitational potential Φ and its spatial derivatives $\partial_i \Phi$ must be computed dynamically via interpolation over a 3D grid (using the Cloud-in-Cell scheme). This operation is not separable from the integrator and must be coupled with the geodesic evolution. Second, this manual implementation is part of a broader strategy to prepare a high-performance version of the code in C++, where full control over memory and operations is required.

The evaluation of the potential gradient $\nabla \Phi$ is performed using central finite differences applied to the grid values. In practice, after computing Φ on the grid (analytically or via

FFT), we compute its derivatives using a second-order centered scheme. These values are then interpolated at each step of the integration using the same CIC scheme, to maintain consistency and avoid artifacts. This allows the geodesic equations to be evolved with gradients that remain smooth.

3.3.1 Evaluation of numerical errors

We will test the integrator on two simple well known Ordinary Differential Equations (ODEs) in physics, in order to check that it works properly. The first one will be the exponential growth,

$$\frac{df}{dt} = f \quad (3.5)$$

and the second one will be the harmonic oscillator,

$$\frac{d^2f}{dt^2} = -f \quad (3.6)$$

The first one is a simple ODE, and the second one is a second order ODE. Both of them have analytical solutions, which will allow us to check that the integrator does return the correct results. We will also check the optimal step size for the integrator, which is such that error is minimized. We can see on the figures 3.5 and 3.6 that the integrator does return the correct results, as the errors are very small, inferior to 10^{-7} for time steps of order 10^{-3} . The convergence plots show that the integrator is optimal for time steps of order 10^{-3} . Time steps higher than that will lead to a divergence of the results, and time steps lower than that will accumulate rounding errors, and will lead to a divergence of the results, as well as a significantly longer computation time.

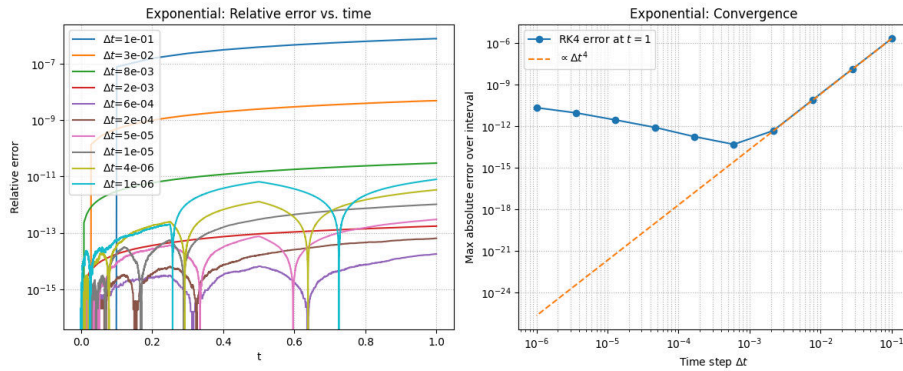


Figure 3.5: Testing the integrator on the exponential growth ODE. On the left, we can see the relative error for different step sizes, and on the right we can see the convergence of the integrator for different step sizes.

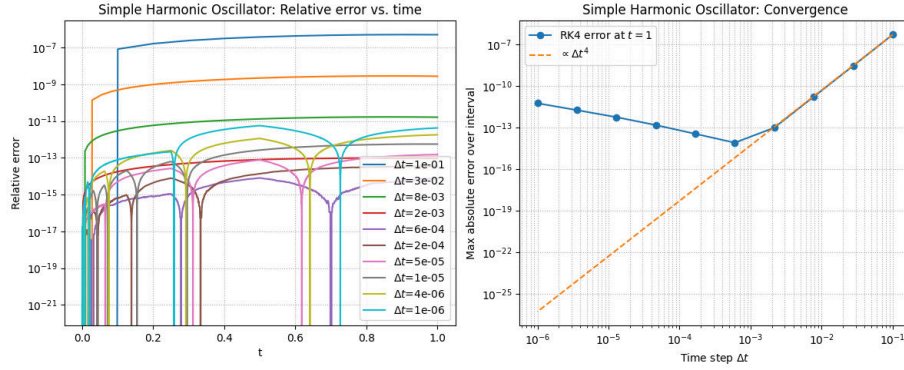


Figure 3.6: Testing the integrator on the harmonic oscillator ODE. On the left, we can see the relative error for different step sizes, and on the right we can see the convergence of the integrator for different step sizes.

To summarize the simulation strategy: we define a grid, fill it with astrophysical objects, compute and interpolate the potential and its derivatives, and integrate the null geodesic equations using a custom RK4 solver. All these steps are designed to be modular and transparent, and to allow a future translation into a high-performance C++ framework. With these tools in place, we can now turn to the first results obtained from this implementation.

CHAPTER 4

Results and interpretation

In this chapter, we will present the first photon trajectories obtained via the previously described methods. Given the cosmological background we are working in — static Universe, stationary potentials, and no peculiar velocities — it can be hard to find physically meaningful results. For example, cosmological redshifts are completely forfeited from our analysis, since there is no expansion of the Universe in our current toy-model. However, the first advantage of those hypothesis is that we can directly observe the photon trajectories with a simple 3D plot. Indeed, since there is no expansion of the Universe, it is meaningful to represent different time slices of the same photon trajectory in a single plot.

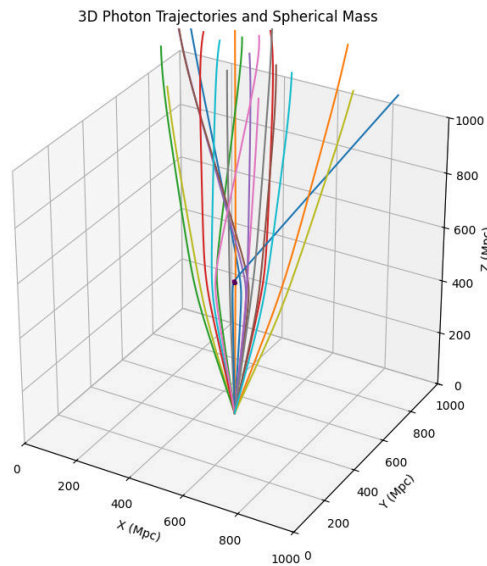


Figure 4.1: Photon trajectories in the presence of a single mass. The mass is represented by the purple sphere, while the photons are all the colored lines that are emitted from an observer located at the center of the xy-plane.

In figure 4.1, we can see the photon trajectories in the presence of a single mass. Here we have taken an absurdly large mass of order $10^{20}M_{\odot}$, in order to be able to see the deformations of trajectories. Let us comment on those deformations, we can see some photons (the blue one going to the right for instance) that take a sharp turn when going near the center of the potential. This shows that a regular grid (we recall that the resolution here is only of $200 \times 200 \times 200$ cells, which is extremely coarse) is very limiting, as we need to question the trajectories that go too close to the center of the grid, especially for very high masses. This is of course only for illustrative purposes, as the mass represented here represents about 2% of the total mass of the observable Universe, concentrated in a radius of about 3 Mpc. For realistic masses (from $10^{13}M_{\odot}$ to $10^{16}M_{\odot}$), the effects of the mass on the photon trajectories would be way more subtle, and not visible in a 3D plot. It would require a more sophisticated analysis relying on other observables, and a good enough number of trajectories to make a satisfactory statistical analysis.

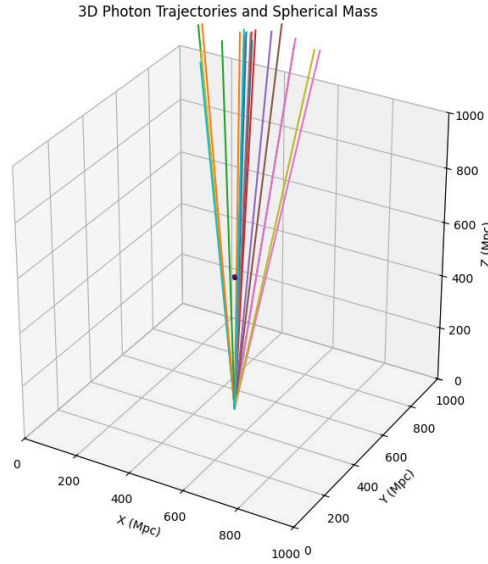


Figure 4.2: Photon trajectories in the presence of a more realistic mass of order $10^{16}M_{\odot}$.

A more realistic mass of order $10^{16}M_{\odot}$ is represented in the figure 4.2. We can see that the geodesics are not as deformed as in the previous case, and we will need to be more careful in the analysis of such trajectories.

One of the main motivations of this work, beyond studying the General Relativistic approach of gravitational lensing, is to study the possible errors that can arise from considering that all halos are spherical. This is a base hypothesis of the current analysis of gravitational lensing, and it is important to understand how much it can affect the results. In order to do so, we will need to study the deformations of the photon trajectories in the presence of a non-spherical mass distribution. This can be done by solving the Poisson equation for a given

non-spherical mass distribution, but for that we need to use tools like Fast Fourier Transforms or multigrid approach. The FFT, as explained in the previous chapter, is a very powerful tool to solve the Poisson equation, but in a cosmological context and in comoving coordinates only, due to the periodic boundary conditions. The multigrid approach allows for a more general approach, but is much more computationally expensive depending on the source function and the grid type [14], and given that the prototype is written in Python, computational resources are limited. Another option would be to approximate mass distributions with a series of summed spherical halos, for which we know the solution to the Poisson equation. This is the approach we will take for now, and we will display some use cases.

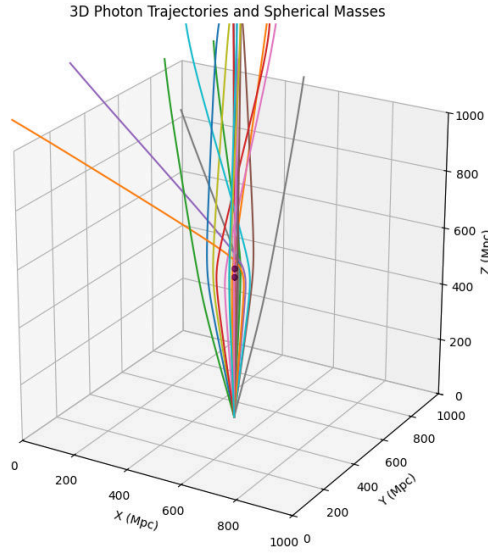


Figure 4.3: Photon trajectories in the presence of two close masses. The masses are equivalent to one mass of order $10^{20}M_{\odot}$, and are located at a distance of around 50 Mpc.

In figure 4.3, we can see the photon trajectories in the presence of two close masses. This way of representing the masses can be considered as a first approximation of an elliptical mass distribution. The total mass is of order $10^{20}M_{\odot}$, still for illustrative purposes. Given the disposition of the halos with respect to the observer, we can already allow ourselves to think that studies of such configurations will be the ones that will matter the most when questioning the thin lens approximation. Indeed, if the mass is distributed differently along the line of sight, we can not expect the same results from the geodesics and from the thin lens approximation.

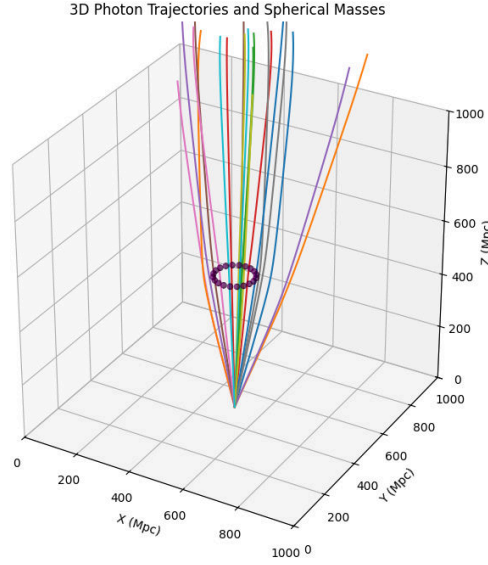


Figure 4.4: Photon trajectories in the presence of a mass distribution that is torus-like, or more familiarly, donut-shaped. The total mass is equivalent to one mass of order $10^{20}M_{\odot}$.

Here we get into the more complicated mass distributions. The donut mass distribution is of course not realistic, but it is a good example of how complex we can get with the mass distributions and the simplicity we want EXCALIBUR to have in handling such cases. We can already see that for three different cases of same total mass, represented in figures 4.1, 4.3 and 4.4, the photon trajectories are very different.

4.1 Time delay measurements

Not studying redshifts, although it is limiting, doesn't mean that there is nothing to study within the constraints we gave ourselves earlier. We will study time delays with the geodesics. To do so, we compare the propagation times for two photons, one subject to geodesic deformations, and one undeflected photon. The following plots are statistics made over 10000 photons. They were propagated on a grid of spatial resolution 5 Mpc, containing a lens of mass $10^{15}M_{\odot}$ and variable radius, to see the influence of the radius on the look and values of the delay map.

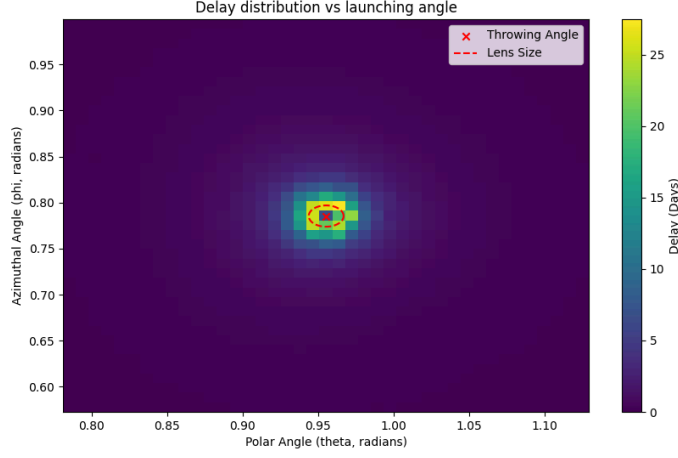


Figure 4.5: Time delay map of photons for a lens with mass $10^{15}M_{\odot}$ and radius 10 Mpc.

We can see that the delay profile from fig. 4.5 follows the circular shape we gave the halo when plugging it into the simulation grid. We are able to measure effects of the order of a day, when the total propagation times are of the order of 1 Gyr. This is already a bit reassuring, although we made a lot of simplifying hypothesis, the results are still compatible with values mentioned in literature [15]. Let's see the influence of radius on the delay map with fig. 4.6

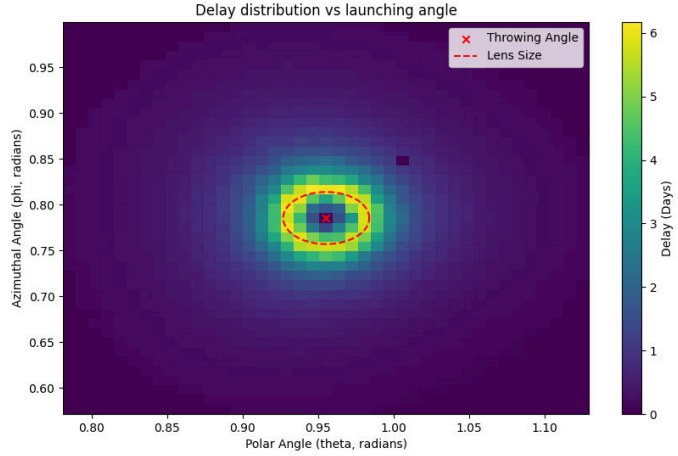


Figure 4.6: Time delay map of photons for a lens with mass $10^{15}M_{\odot}$ and radius 25 Mpc.

We notice that in fig. 4.6 the maximum delay is reduced, the ring is wider, but the shape is still precisely aligned with the known shape of the halo inserted in the simulation. We can discuss the photons that go through the red dashed-line circle, as those photons would technically be inside the halo. This opens astrophysical questions about the propagation of light in galactic medium, so let's not delve into those considerations. However, it is still interesting to see that the maximum of the time-delay is aligned precisely on the halo's shape.

Just for showcasing, we also studied the time delay map resulting from a similar configuration as fig. 4.4, except the photons leave from the origin instead of the center of the Oxy plane.

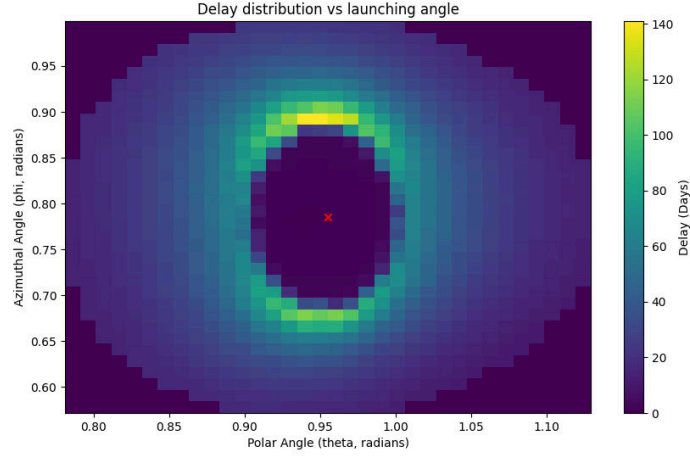


Figure 4.7: Time delay map of photons for a donut shaped lens with total mass $10^{16}M_{\odot}$ and radius 40 Mpc.

This time, it is meaningful to consider the photons that go through the central ring, as there is no mass present there. Those photons exhibit very low delays, of about a day up to 20 days when getting close to a side of the donut of fig. 4.7. On a more serious note, let us see that in a massless configuration, we correctly retrieve what we expect to retrieve : a kind of noise map, determining our numerical accuracy for the estimation of time delays.

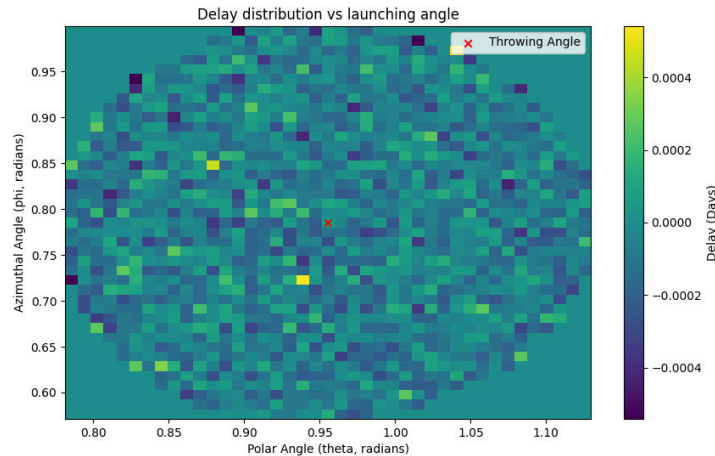


Figure 4.8: Time delay map of photons for an empty grid.

Thanks to this map 4.8, we see that the delay is estimated up to (in absolute value) 0.0004

days $\approx$ 30 seconds.

4.2 Ongoing work

Same as before, for showcasing reasons, we show here the first attempts at finding a calibration of mass/radius vs delay, to retrieve an estimation of the mass and shape thanks to the measurement of the time delay map.

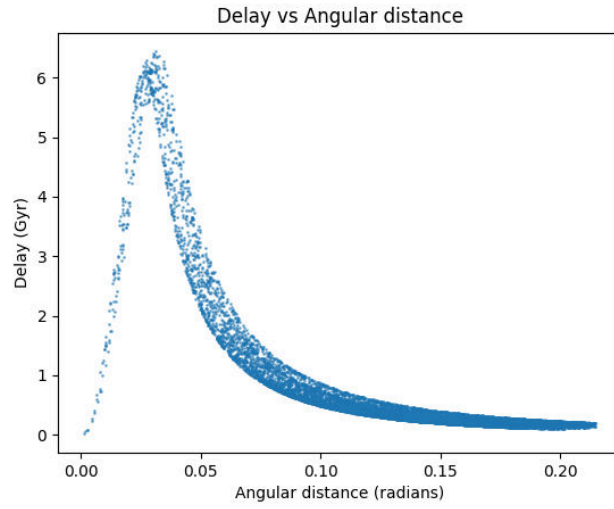


Figure 4.9: Estimating the radius of a lens with known mass $10^{15}M_{\odot}$ and radius 25 Mpc thanks to delay measurements.

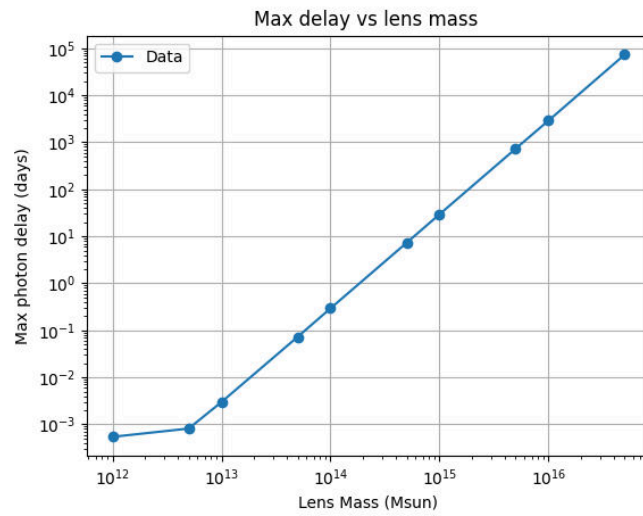


Figure 4.10: Calibration of the mass/delay relation.

Figures 4.10 and 4.9 are very preliminary, errors on the delays haven't even been estimated yet, due to time constraints. However, we can see that there seems to be a relation between the three different quantities: mass and radius of the lens and max delay measured. Ignoring the leftmost point of fig. 4.10, the curve seems to follow a power-law, more specifically M^2 , with M the mass of the lens.

CHAPTER 5

Perspectives

In this short chapter, we will discuss the perspectives of this work. We will first go over the possible theoretical improvements that can be made to the current state of the work, along with possible interesting applications for refining some theoretical aspects. We will then look at the possible improvements that can be made to the code, and how we can improve the current state of the code to make it more efficient and easier to use. Finally, we will talk about the possible applications of EXCALIBUR to the study of gravitational lensing, and how we will be able to use it to bring new insights into the field.

5.1 Theoretical perspectives

The current framework relies on a linear perturbative expansion of the FLRW metric and on simplifying assumptions such as the Newtonian limit and time-independence of the gravitational potential. A natural extension would be to relax these assumptions. Including vector and tensor perturbations — particularly relevant for incorporating effects of gravitational waves — and allowing the potential $\Phi(x, t)$ to evolve with time would bring the model closer to a realistic cosmological setting. Beyond linear order, exploring non-perturbative or higher-order corrections could also shed light on strongly non-linear environments, such as the cores of galaxy clusters or merging structures.

5.2 Numerical perspectives

The current implementation is based on a regular grid, which offers simplicity but becomes memory-intensive at high resolutions. A major planned upgrade is the development of an Adaptive Mesh Refinement (AMR) structure. AMR would enable dynamic resolution enhancement in high-density regions, optimizing the balance between precision and computational cost. This transition would also require reworking gradient interpolation schemes and ensuring numerical stability of the geodesic integration within a non-uniform mesh.

The implementation of AMR is a promising avenue to better capture the steep gradients and fine structures near massive objects. It is particularly relevant for modeling the small-scale features of strong lensing, such as critical lines and caustics. By increasing the resolution

locally while keeping memory usage low globally, AMR grids offer a scalable solution for future simulations involving complex, multi-halo environments. Particular attention must be given to ensuring continuity and differentiability of the interpolated fields across refinement levels.

Translating the project to C++ will also allow to propagate billions of photons and obtain realistic images in reasonable times. We also intend to include different other types of numerical integrators, such as a fourth order Yoshida scheme or an adaptive RK45 scheme, to evaluate numerical errors more precisely. A natural course of improvement will also be to implement a TSC (Triangular Shape Cloud) interpolation instead of CIC.

5.3 Observational perspectives

A long-term objective of this project is to produce mock convergence and shear maps directly from simulated photon trajectories. Our first step is to establish a consistent framework that simultaneously models both weak and strong lensing effects.

We will then incorporate instrumental effects such as the point spread function (PSF) to enable a realistic comparison between these simulations and observational data from large upcoming surveys like LSST and Euclid. While the Nancy-Grace-Roman Space Telescope is often mentioned, its data are not yet available and may never be, so our focus remains on LSST, whose data will arrive very soon.

Careful comparison between data and simulations, once these instrumental effects are accounted for, will allow us to extract meaningful astrophysical information, including the masses of galaxy clusters and the distribution of dark matter.

An important aspect of our framework is the unified treatment of multiple relativistic effects influencing photon paths: gravitational lensing, the integrated Sachs-Wolfe effect, peculiar velocities, among others. This will enable us to evaluate the relative importance of these effects—for instance, at given redshift, to what extent peculiar velocity effects can be neglected compared to gravitational redshift.

Furthermore, LSST will deliver an unprecedented sample of strong lensing events, making possible statistical studies of these phenomena for the first time.

Finally, this framework also opens the door to inverting the lensing problem: reconstructing mass distributions from lensing observations. By training machine learning models on simulated data, we aim to develop fast and reliable mass inference tools applicable to large lens catalogs—potentially providing new insights into the nature of dark matter.

CHAPTER 6

Conclusion and Future Work

This Master’s thesis set out to explore a unified approach to gravitational lensing by directly integrating null geodesics in a perturbed FLRW spacetime, with the goal of bridging the conceptual and methodological gap between weak and strong lensing regimes. This task, though ambitious, is well-motivated by the upcoming flood of high-quality data from next-generation surveys like LSST and Euclid, which will require consistent modeling tools capable of handling the continuum between both lensing regimes.

On the theoretical side, the work has revisited the fundamental aspects of photon propagation in curved spacetime and has translated these into a numerical framework based on General Relativity, using linear scalar perturbations. This approach offers a clear geometric interpretation of lensing phenomena, avoids the usual approximations such as the Born approximation, thin lens model, and single-plane hypothesis, and naturally incorporates relativistic effects beyond lensing alone. On the numerical side, a prototype code — EXCALIBUR — was developed to simulate photon trajectories in 3D space, compute gravitational potentials from arbitrary mass distributions, and visualize the resulting light paths. The modular design of the code allows for easy upgrades and extension to more complex configurations. It also sets the stage for future implementations in C++ to improve computational efficiency.

Several case studies were presented, from single and dual-mass setups to toroidal (donut-shaped) mass configurations. These simulations clearly demonstrated how the same total mass, distributed differently in space, leads to qualitatively distinct deflection patterns — a result that highlights the importance of moving beyond spherically symmetric halo models in lensing studies.

While the framework is still in its early stages, it already opens up several promising directions. These include the development of AMR grids, the inclusion of time-varying and vector/tensor perturbations, and the eventual coupling with machine learning techniques for mass inference. Importantly, as modeling uncertainties will dominate future analyses, this work tackles a core modeling challenge directly. All of these will be addressed in the continuation of this project, which is expected to evolve into a PhD thesis. In summary, this work lays the foundation for a unified, flexible, and physically grounded tool to study light propagation at cosmological scales. It demonstrates the feasibility and advantages of treating light propagation through realistic, inhomogeneous Universes using general relativistic principles. With further developments, this approach could significantly enhance our

ability to interpret future cosmological data and improve our understanding of the large-scale distribution of matter in the Universe.

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List of Figures

1.1	Examples of images from previous galaxy surveys : (a) shows the density of background galaxies LSST will capture for weak lensing analysis. (b) shows a spectacular strong lensing event caused by a massive foreground galaxy cluster.	4
3.1	Visualizing the cone of observation mentioned earlier. The observer is located at the origin, and the cone is defined by the angle of observation and the aperture angle. The cone is represented by the blue points.	16
3.2	Slice of the potential (left) and density (right) of a single halo, with a mass of $10^{13}M_{\odot}$ and a radius of 1 Mpc, at coordinate $z=0.5$. The potential is represented in $\text{Mpc}^2/\text{Gyr}^2$, and the density is represented in Msun/Mpc^3	18
3.3	Comparing two slices of the potential, one computed using the FFT method (left) and one computed using the analytical solution (right). The potential is represented in $\text{Mpc}^2/\text{Gyr}^2$	19
3.4	Relative difference between the potential computed using the FFT method and the potential computed using the analytical solution. The x and y axes are in Mpc. The relative difference is defined as $\frac{\phi_{fft}-\phi_{analytic}}{\phi_{analytic}}$	19
3.5	Testing the integrator on the exponential growth ODE. On the left, we can see the relative error for different step sizes, and on the right we can see the convergence of the integrator for different step sizes.	22
3.6	Testing the integrator on the harmonic oscillator ODE. On the left, we can see the relative error for different step sizes, and on the right we can see the convergence of the integrator for different step sizes.	23
4.1	Photon trajectories in the presence of a single mass. The mass is represented by the purple sphere, while the photons are all the colored lines that are emitted from an observer located at the center of the xy-plane.	24
4.2	Photon trajectories in the presence of a more realistic mass of order $10^{16}M_{\odot}$. .	25
4.3	Photon trajectories in the presence of two close masses. The masses are equivalent to one mass of order $10^{20}M_{\odot}$, and are located at a distance of around 50 Mpc.	26
4.4	Photon trajectories in the presence of a mass distribution that is torus-like, or more familiarly, donut-shaped. The total mass is equivalent to one mass of order $10^{20}M_{\odot}$	27
4.5	Time delay map of photons for a lens with mass $10^{15}M_{\odot}$ and radius 10 Mpc. .	28
4.6	Time delay map of photons for a lens with mass $10^{15}M_{\odot}$ and radius 25 Mpc. .	28

List of Figures

4.7	Time delay map of photons for a donut shaped lens with total mass $10^{16}M_{\odot}$ and radius 40 Mpc.	29
4.8	Time delay map of photons for an empty grid.	29
4.9	Estimating the radius of a lens with known mass $10^{15}M_{\odot}$ and radius 25 Mpc thanks to delay measurements.	30
4.10	Calibration of the mass/delay relation.	30

List of Tables

- 2.1 The perturbed Christoffel symbols in a $k = 0$ spacetime, with all vector and tensor perturbations set to zero. The terms Φ and Ψ are the scalar perturbations. The terms $\dot{a}$ and a are the time derivative of the scale factor and the scale factor respectively. 9